

Shreya Denre

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Professional Summary

Third-year Computer Science undergraduate with experience in building AI-enabled web applications and scalable backend systems. Strong in JavaScript, React, and Node.js, with hands-on experience integrating APIs and developing intelligent features using LLMs and AI tools. Proven ability to design end-to-end systems that combine frontend, backend, and data-driven insights for real-world applications.

Education

Heritage Institute of Technology, Kolkata

Bachelor of Technology in Computer Science & Engineering

Sept 2023 – Aug 2027

- **CGPA:** 9.6 (Ranked in top 20%: [Certificate](#))
- **ICSE:** 96% | **ISC:** 96% | **WBJEE Rank:** 4155 | Gate Qualified

Work Experience

Data Science Intern | ISI-Kolkata [Certificate](#)

Jan,2026-March, 2026

- Built an end-to-end **AI-powered video analytics pipeline** for traffic monitoring using detection, tracking, and segmentation models
- Designed data processing workflows and applied predictive models for traffic flow analysis and congestion insights.

Machine Learning Intern | Code-X-Novas [Certificate](#)

July,2025-August, 2025

- Implemented deep learning models for hyperspectral image classification using attention mechanisms and TopoMorph RG Convolutional Networks for enhanced spatial-spectral feature extraction
- Achieved 98%+ accuracy through architecture optimization and hyperparameter tuning; validated performance using evaluation metrics and t-SNE visualization

Research Area

Emotion Detection from Social Media Texts by NLP (ongoing research)

Tech Stack: Python, PyTorch, Hugging Face Transformers, Scikit-learn

- Built an NLP pipeline using transformer models (DistilBERT) for emotion detection in noisy and code-mixed social media text
- Designed preprocessing and evaluation workflows to handle ambiguity, class imbalance, and multilingual data (Hinglish/Benglish)
- Applied model analysis techniques to improve detection of subtle signals such as sarcasm and minority emotion classes

Projects

CureConnect- AI Powered Healthcare Platform [Live link](#)

MERN, Cloudinary, Python, Flask, Scikit-Learn, Gemini API, Streamlit

- Built a **full-stack** healthcare platform with **doctor registration & appointment booking** and **ambulance driver onboarding with verification system**
- Implemented **real-time ambulance booking with live location tracking using OpenStreetMap (OSM)** for emergency coordination
- Developed AI-driven features including **Gemini-based prescription analysis** and **symptom-based disease prediction** with suggested remedies, integrated via secure REST APIs with JWT authentication and RBAC.

NrityaLens- AI-Powered Bharatanatyam Learning Platform [Live link](#)

MERN, Python, Flask, FastAPI, Gemini API, LangGraph

- Developed a CNN-based **mudra detection system** (trained on 30k+ dataset) and integrated it with backend APIs for real-time prediction
- Built a **full-stack** platform with image/camera-based **mudra detection, digital mudra library, and AI assistant** for interactive learning
- Implemented features like **mudra assessment, progress analytics, and group learning workflows**, enhancing user engagement and learning insights.

Nexus.ai- AI Career Platform [Demo](#) | **MERN, Clerk Auth, Next.js, NeonDB, Inngest, Tailwind CSS, Groq API, LLaMa**

- Built a full-stack AI-powered platform with **event-driven multi-agent workflows** for **resume analysis, job matching, roadmap generation, quiz and intelligent chatbot**.

- Implemented **structured AI output handling with schema validation**, ensuring reliable backend-to-frontend data flow and consistent responses
- Developed features including **resume parsing, skill gap analysis, AI-generated interview questions, and personalized learning roadmaps**
- Integrated **real-time contest tracking with Google Calendar**, driving user engagement and workflow automation (50+ users)

GreenCircuit – AI-Powered E-Waste Management Platform [↗](#)

[Demo](#)

React.js, Node.js, Express.js, MongoDB, Gemini API

- Built a full-stack platform connecting sellers, buyers, and recyclers with multi-role dashboards, dynamic item listing, pickup scheduling and RESTful APIs for end-to-end item management.
- Integrated **AI-based classification** to categorize e-waste from images/descriptions, improving decision-making and automation
- Implemented **QR code-based lifecycle tracking** (listing → verification → transfer → recycling) with real-time scanning, status updates, route optimization using OSM maps, and unique serial number tracking

Auralix.ai- Meeting Intelligence Tool [↗](#)

[Demo](#)

Python, FastAPI, Flask, Whisper, Gemini API, Notion API, Slack API, GitHub API, Chrome Extension

- Built a speech-to-text pipeline using **Whisper API** to convert unstructured audio into structured text for NLP analysis.
- Developed a **real-time Chrome extension** for automated meeting capture and transcript generation, integrating transcription outputs with NotionDB and Slack to support meeting summarization and task extraction.
- Applied **automation** and workflow optimization with **CI/CD** and **Cron** jobs to ensure reproducibility during experimental evaluation.

Technical Skills

Languages: C++, Python, JavaScript, TypeScript, Java

Core CS: Data Structures, Algorithms, OOP, Operating Systems, Cloud computing, DBMS, Software Engineering

Frontend: React.js, Next.js, Tailwind CSS, Angular

Backend: Node.js, Express.js, REST APIs, Third-party API integration

Databases: MongoDB, PostgreSQL, MySQL, DrizzleORM, NeonDB, SQLite3,

AI/ML: PyTorch, Pandas, Numpy, scikit-learn, Hugging Face, NLP, Deep Learning, LangChain, Embeddings, OpenCV

Tools: Git, GitHub, Docker & AWS Basics, CI/CD Pipelines, Vercel, Render, Postman

Achievements & Certifications

- Completed 6-week Summer Analytics 2024 course ([Data Science & ML](#)) by Consulting & Analytics Club, IIT Guwahati 
- **NPTEL** courses on Machine Learning, Neural Networks for Computer Vision and NLP, and Introduction to LLMs.
- **Second Runner-Up**, Hack4Bengal 4.0 — Eastern India's largest hackathon, 36-hour build [↗](#)
- Nominated for Smart India Hackathon (SIH) 2025
- Participated in IEM-DIVERSION 2K25(Inter-college hackathon organised by IEEE Student Branch, Institute of Engineering and Management Kolkata, WB): [Certificate](#) [↗](#)
- Participated in HexaFalls 2025 (organised by JIS University).
- Participated in HackOdisha5.0: [Certificate](#) [↗](#)